MME OSPE

Station 1

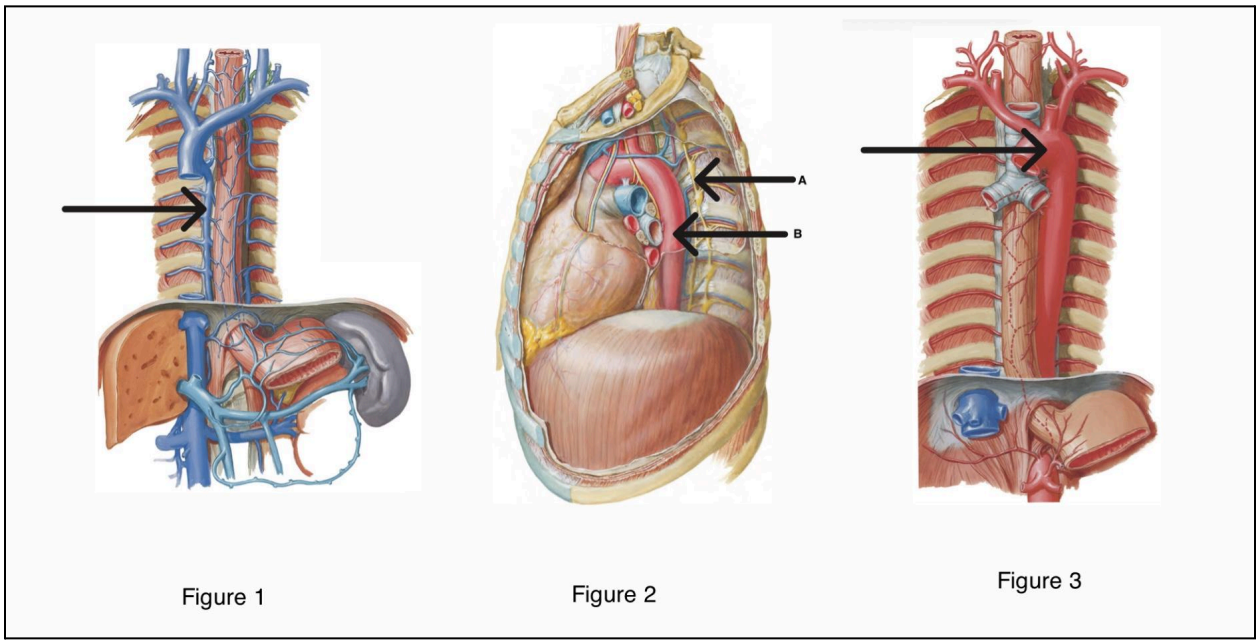


Figure 1 Figure 2 Figure 3

1. Identify the structure pointed at in figure 1, where does it terminate
    - Azygous vein system, superior vena cava at the plane of Ludwig

2. Identify the structures A and B in figure 2
    - A: sympathetic chain
    - B: descending thoracic aorta

3. Identify the structure in figure 3, name one of its branches
    - Arch of aorta, brachiocephalic trunk

Station 2

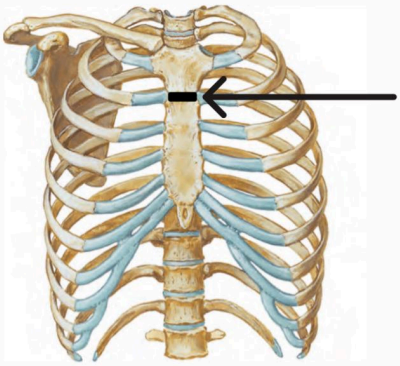


Figure 1

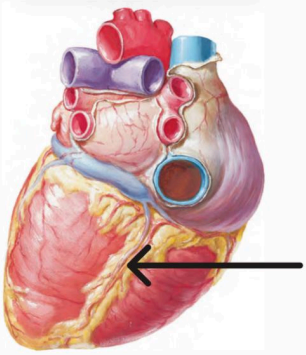


Figure 2

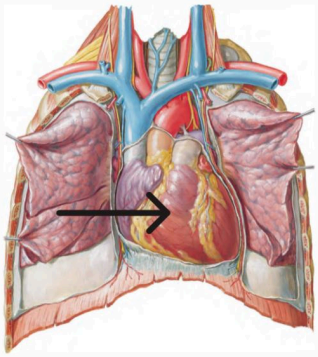


Figure 3

1. Identify the appointed black line in figure 1, mention two of its significance
    - Sternal angle of Louis
    - The arch of aorta starts and ends, trachea bifurcates into left and right main bronchi, second rib articulates directly and helps in rib counting for physical examination

2. Identify the groove in figure 2 and two of its components
    - Posterior inter ventricular groove lodging posterior descending artery and middle cardiac vein

3. Identify the heart chamber in figure 3, its embryological origin and one vessel arising from it
    - Right ventricle
    - Rough part (traneculae carnae) arises from the primitive ventricular chambers and smooth part (infundibulum) arises from bulbous cordis
    - Pulmonary trunk

Station 3

(Scenario) Mr. R, 62 years old, is suffering from hypertension and angina pectoris

1. Prescribe and include directions for use

   Rx
   Tab. Glyceryl trinitrate (GTN) 300 mcg sublingually for acute angina attacks
   Tab. Metoprolol tartrate 25 mg twice a day for 1 month
   Tab. Enalapril 5 mg once a day for 1 month
   Review after one month

2. Why did you choose these therapeutic options for this patient

   - Glyceryl Trinitrate is a nitrate that works by dilating blood vessels (vasodilation), which reduces the workload on the heart and improves blood flow to the myocardium. This helps alleviate angina symptoms quickly.
   - Metoprolol Tartrate: controlling heart rate and reducing myocardial oxygen demand, which helps alleviate angina symptoms. Also helps to lower blood pressure, contributing to overall cardiovascular health.
   - ACE Inhibitor: Helps manage hypertension by relaxing blood vessels, which lowers blood pressure and reduces strain on the heart. Provides benefits for patients with potential heart disease, reducing the risk of cardiovascular events.

Station 4

Write the appropriate heart sound for each of the following, one heart sound can be used more than once

Rapid passive ventricular filling during early diastole	S3
After the onset of QRS complex on ECG	S1
Atria contract against non complaint “stiff” ventricles	S4
At start of isovolumetric contraction	S1
Immediately after T wave on ECG	S2